

From first principles to the browser: new instruments for computational historical linguistics

Computational historical linguistics has borrowed its models and its epistemology from biology (Gray & Atkinson 2003), without really addressing questions on whether its objects behave like theirs or whether the analogies hold. It assumes much of what it sets out to find: cognacy is reduced to curated binary lists in selected vocabulary, the character model is inherited from morphological phylogenetics (Lewis 2001), trees are the generative model rather than a hypothesis, direction is settled by likelihood majority, and regularity is a gate rather than a measurement. Criticisms range from the discard of phylogenetic signal carried by the word forms themselves (Goldstein et al. 2026) to the demonstration that common models fail adequacy tests against the data they are fitted to (King 2026).

In this software demonstration I will discuss an alternative framework, built on a constraint: nothing the analysis is meant to establish may enter it as data. Where the evidence does not separate the alternatives, the framework says so, prompting the expert for more data. The core evidence is phonological, that is, systematic correspondences over a given lexicon, not cognacy over basic-vocabulary slots. Correspondences are stated in both directions and weighted asymmetrically (Kümmel 2007), so direction is inferred rather than settled by majority. Regularity is not assumed but measured against a baseline, and confidence levels are carried throughout the analyses.

The demonstration will focus on *regulae*, a tool for detecting systematic sound correspondences with their residue, and on *merkmal*, the phonological feature library underneath it. *regulae* discovers its conditioning environments instead of being handed them. It can, among other things, find long-range triggers such as in umlaut and harmony, and cross-dimensional rules where voicing on one lect's initial predicts tone on another's vowel. It also declines when there is nothing there: given wordlists with no shared history, no conditioned correspondence survives the search. It has been run over several dozen published wordlists (List et al. 2022), from Indo-European to Pama-Nyungan. Like all software in this framework, it is implemented as a dependency-free library that can be run in a local web page (no server or installation required), designed to be functional for many years.

References

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